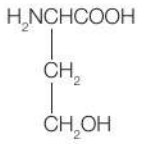
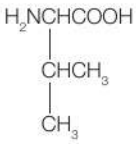
15 EXAM PRACTICE

1 Which amino acid contains two chiral carbon atoms?

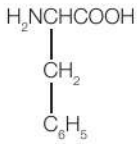
A



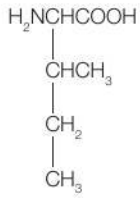
B



C



D



[1]

(Total for Question 1 = 1 mark)

2 The same carboxylic acid is obtained either by the hydrolysis of a nitrile ***P*** or by the oxidation of an alcohol ***Q***.

Which pair could be ***P*** and ***Q***?

	P	***Q***
A	CH_3CH_2CN	CH_3CH_2OH
B	$(CH_3)_2CHCN$	$(CH_3)_3COH$
C	$C_6H_5CH(CH_3)CN$	$C_6H_5CH_2CH(OH)CH_3$
D	$C_6H_5CH_2CN$	$C_6H_5CH_2CH_2OH$

[1]

(Total for Question 2 = 1 mark)

3 Which pair of reactants produces an ester with the formula $C_3H_7COOC_2H_5$?

A C_2H_5Cl and C_3H_7COOH

B C_2H_5OH and C_3H_7COOH

C C_3H_7OH and C_2H_5COCl

D C_3H_7OH and C_2H_5COOH [1]

(Total for Question 3 = 1 mark)

4 One of the constituents of beeswax has the formula $CH_3(CH_2)_{24}CO_2(CH_2)_{29}CH_3$.

What are the products of its alkaline hydrolysis?

A $CH_3(CH_2)_{24}COOH$ and $CH_3(CH_2)_{29}O^-$

B $CH_3(CH_2)_{24}COOH$ and $CH_3(CH_2)_{29}OH$

C $CH_3(CH_2)_{24}COO^-$ and $CH_3(CH_2)_{29}OH$

D $CH_3(CH_2)_{24}COO^-$ and $CH_3(CH_2)_{29}O^-$ [1]

(Total for Question 4 = 1 mark)

5 An organic compound has the following properties:

- it gives a positive tri-iodomethane (iodoform) test
- it gives a yellow precipitate with 2,4-dinitrophenylhydrazine
- it does not react with either Tollens' reagent or Fehling's solution

Which compound gives these results?

A CH_3CHO

B CH_3CH_2OH

C $CH_3CH_2COCH_3$

D $CH_3CH_2CH_2CHO$ [1]

(Total for Question 5 = 1 mark)

6 What are the principal inorganic and organic products when propanal reacts with Tollens' reagent?

A Ag and CH_3CH_2COOH

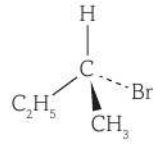
B Ag and $CH_3CH_2COO^-$

C Ag_2O and CH_3CH_2COOH

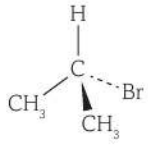
D Ag_2O and $CH_3CH_2COO^-$ [1]

(Total for Question 6 = 1 mark)

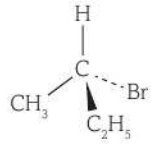
7 The structures of three organic compounds, **A**, **B** and **C**, are shown.



A



B



C

(a) Explain which two of these compounds are enantiomers. [2]

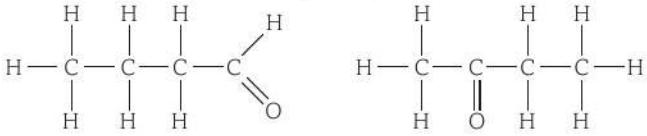
(b) Explain why a mixture containing equal numbers of these enantiomer molecules cannot be distinguished from the other compound of the three in a polarimeter. [2]

(c) Compound **A** reacts with aqueous sodium hydroxide.

Explain how measurements of the optical activity of compound **A** and the organic product of the reaction can be used to confirm whether the reaction occurs by an S_N1 mechanism or an S_N2 mechanism. [6]

(Total for Question 7 = 10 marks)

8 The structures of two carbonyl compounds are:



(a) State why both compounds are soluble in water. [1]

(b) (i) Name the mechanism of the reaction between each of the compounds and hydrogen cyanide. [1]

(ii) Both compounds react with 2,4-dinitrophenylhydrazine.

Describe how the organic product of each of these reactions could be used to identify the carbonyl compounds. [2]

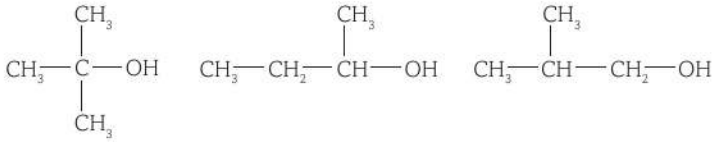
(c) Give the structure of the organic product of the reaction of each compound with lithium tetrahydridoaluminate, followed by water. [2]

(d) Only one of the two compounds reacts on warming with Tollens' reagent.

Write an equation for the reaction that occurs, using [O] to represent Tollens' reagent, and state the observations that will be made when a positive result is obtained. [2]

(Total for Question 8 = 8 marks)

9 (a) The structures of three alcohols are:

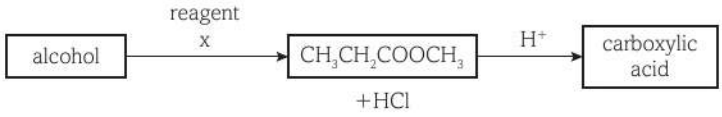


D **E** **F**

Explain which of these alcohols reacts on warming with acidified potassium dichromate(VI) to form a carboxylic acid. [3]

(b) Draw the displayed formula of the nitrile that can be hydrolysed to the carboxylic acid with the formula $(CH_3)_2CHCH_2COOH$. [1]

(c) The diagram summarises a sequence of two reactions.



(i) Deduce the name of the alcohol. [1]

(ii) Explain how the homologous series to which reagent X belongs can be deduced from the diagram. [2]

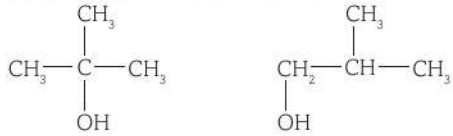
(iii) Deduce the structural formula of the carboxylic acid. [1]

(iv) The yield of carboxylic acid in the second step can be improved by using a different reagent before the acid in a two-step process.

Write an equation for each of these steps. [2]

(Total for Question 9 = 10 marks)

10 The structures of two compounds are:



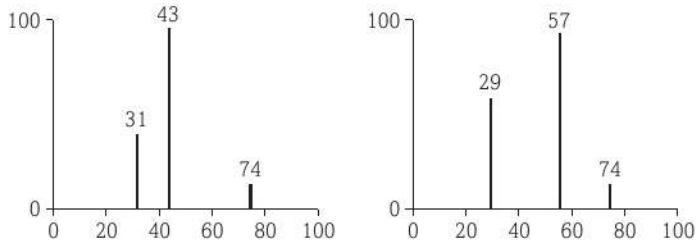
(a) Explain why infrared spectroscopy is not used to distinguish these two compounds. [2]

(b) Explain how the number of peaks in the ^{13}C NMR spectra of these compounds can be used to identify which spectrum is obtained from each compound. [4]

(c) One of these compounds has a ^1H NMR spectrum that shows only two singlets.

Explain which compound can be identified from this information. [3]

(d) A simplified mass spectrum of each of the compounds is shown below.



Explain how this information can be used to identify each compound. [6]

(Total for Question 10 = 15 marks)